

Tutorial 8 (SOLUTIONS)

NOTE THAT THERE ARE OTHER POSSIBLE SOLUTIONS TOO...

Question 1

A	B	Q	Q ⁺
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	0

(i): Characteristic Table

Q	Q ⁺	D
0	0	0
0	1	1
1	0	0
1	1	1

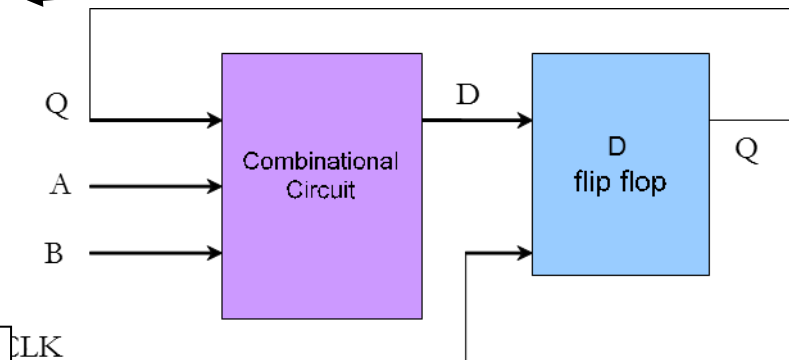
Excitation Table

A	B	Q	Q ⁺	D
0	0	0	1	1
0	0	1	1	1
0	1	0	0	0
0	1	1	1	1
1	0	0	1	1
1	0	1	0	0
1	1	0	0	0
1	1	1	0	0

Combined Table

A	B	Q	D
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	0

(iii): Combinational circuit truth table

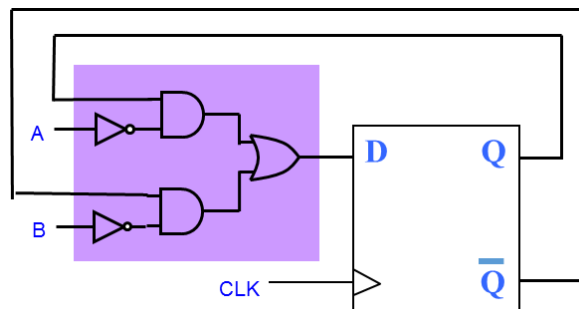


(ii): Functional Block Diagram

e: Use Karnaugh Maps to find logic expressions of D

$$D = \overline{B}\overline{Q} + \overline{A}Q$$

		Q	
		0	1
AB	00	1	1
	01	0	1
	11	0	0
	10	1	0



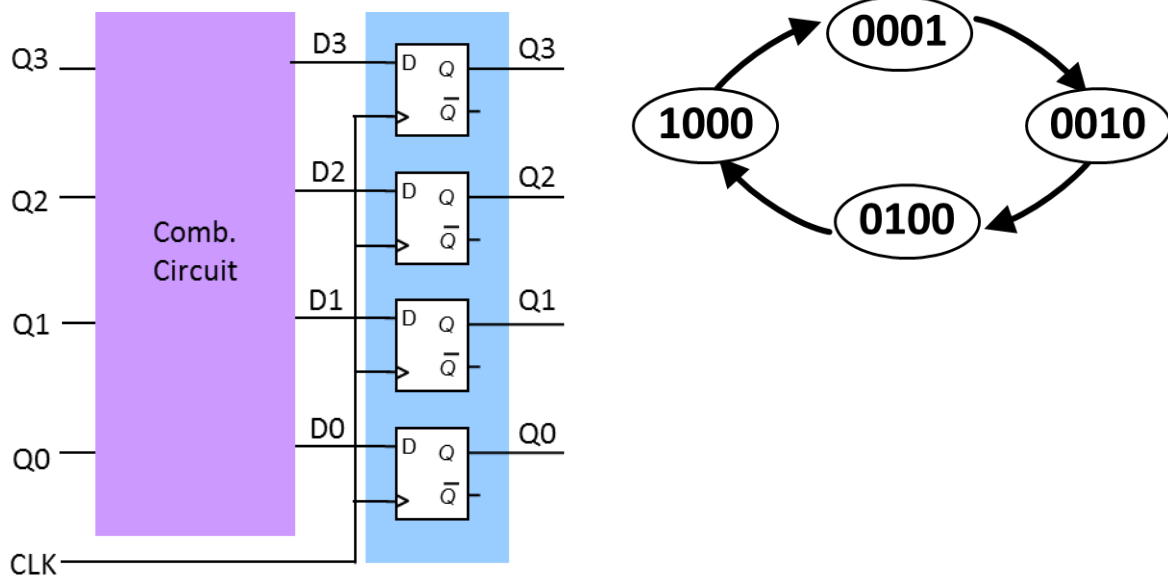
```

module abff (input clk, a, b, output reg q);
  wire d;
  always @ (posedge clk) begin
    q <= d;
  end
  assign d = (~b & ~q) | (~a & q);
endmodule

```

Question 2

1) Functional Block Diagram



2) Next State Table

Current State				Next State			
Q3	Q2	Q1	Q0	Q3+/D3	Q2+/D2	Q1+/D1	Q0+/D0
0	0	0	1	0	0	1	0
0	0	1	0	0	1	0	0
0	1	0	0	1	0	0	0
1	0	0	0	0	0	0	1
Other States				X	X	X	X

3) Combinational Circuit

Q_1Q_0		Q_3+ Kmap			
Q_3Q_2		00	01	11	10
00		X	0	X	0
01		1	X	X	X
11		X	X	X	X
10		0	X	X	X

$Q_3+ = D_3 = Q_2$

Q_1Q_0		Q_2+ Kmap			
Q_3Q_2		00	01	11	10
00		X	0	X	1
01		0	X	X	X
11		X	X	X	X
10		0	X	X	X

$Q_2+ = Q_1$

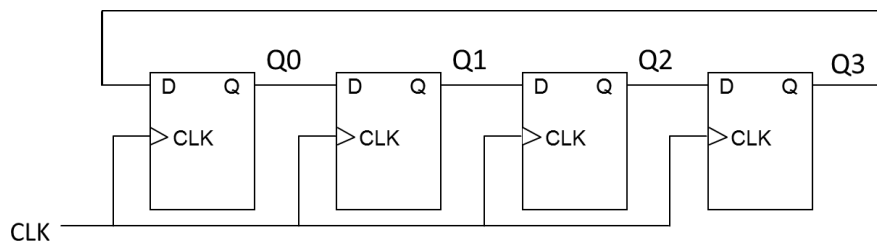
Q_1Q_0		$Q1+ \text{ Kmap}$			
Q_3Q_2		00	01	11	10
00		X	1	X	0
01		0	X	X	X
11		X	X	X	X
10		0	X	X	X

$$Q1+ = Q0$$

Q_1Q_0		$Q0+ \text{ Kmap}$			
Q_3Q_2		00	01	11	10
00		X	0	X	0
01		0	X	X	X
11		X	X	X	X
10		1	X	X	X

$$Q0+ = Q3$$

4) Circuit Realization



As can be seen above, the circuit realization does not require any additional logic gates. Due to the way the flip flops are connected, the bits in the counter *shift* to the right every active clock edge. This is illustrated as below.

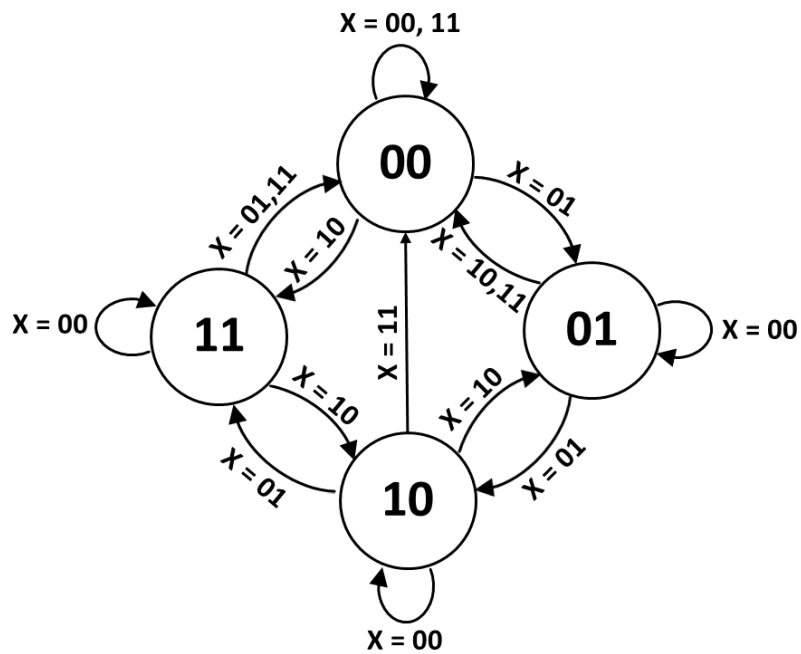
Q0	Q1	Q2	Q3
0	0	0	1
1	0	0	0
0	1	0	0
0	0	1	0
0	0	0	1

However, due to the simple circuit, a tradeoff is that the counter needs to be initialized to a value of 0001, 0010, 0100 or 1000 in order to operate. If the counter is initialized to other values instead (for example 0000), it would not behave as a 1,2,4,8... counter.

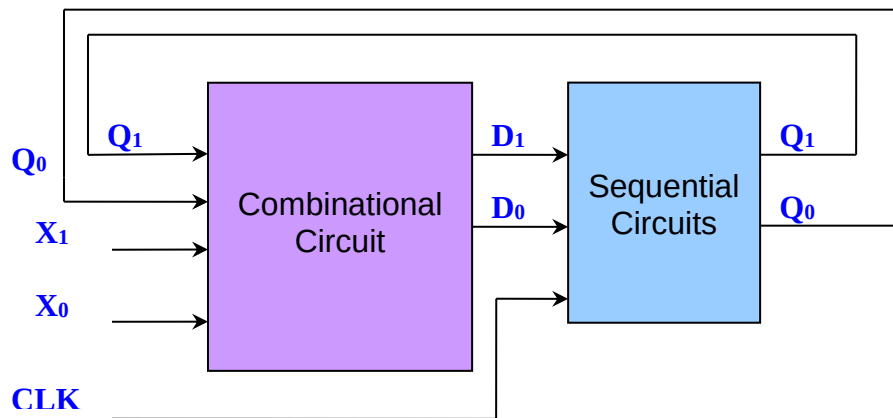
This circuit is also known as a ring counter, or a shift register (where the last stage connected to the first).

Question 3

State Diagram



4 mode 2-bit counter



The next state table is:

X_1	X_0	Q_1	Q_0	Q_1^+ $/D_1$	Q_0^+ $/D_0$
0	0	0	0	0	0
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	1	1
0	1	0	0	0	1
0	1	0	1	1	0
0	1	1	0	1	1
0	1	1	1	0	0
1	0	0	0	1	1
1	0	0	1	0	0
1	0	1	0	0	1
1	0	1	1	1	0
1	1	0	0	0	0
1	1	0	1	0	0
1	1	1	0	0	0
1	1	1	1	0	0

Q_1Q_0	00	01	11	10
X_1X_0				
00	0	1	1	0
01	1	0	0	1
11	0	0	0	0
10	1	0	0	1

$$D_0 = Q_0 \bar{X}_1 \bar{X}_0 + \bar{X}_1 X_0 \bar{Q}_0 + X_1 \bar{X}_0 \bar{Q}_0$$

Q_1Q_0	00	01	11	10
X_1X_0				
00	0	0	1	1
01	0	1	0	1
11	0	0	0	0
10	1	0	1	0

$$D_1 = \bar{Q}_1 Q_0 \bar{X}_1 X_0 + \bar{Q}_1 \bar{Q}_0 X_1 \bar{X}_0 + \bar{X}_1 Q_1 \bar{Q}_0 + Q_1 Q_0 \bar{X}_0$$

